ZISAR: a Zero-Inflated Spatial Autoregressive Model for Count Data

Spillovers, excess zeros and overdispersion in one likelihood

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Where this is going: real data, York, spring 2027

The visit

- April–May 2027, **Centre for Health Economics, University of York**, with **Rita Santos** (Senior Research Fellow; spatial econometrics of health systems, urgent and emergency care).
- Goal: take ZISAR from the Monte Carlo to English NHS data.

The data (details to be agreed with CHE)

- **Emergency-department attendances counted by small area** and period — the same structure as the maps in this talk.
- Per area: resident population and age structure, deprivation and income, distance to the nearest ED, GP supply.
- Area boundaries and centroids, from which **W** is built.

Why it is a ZISAR problem

- Zeros: small areas, short periods and specific causes often record none — and some of those zeros are structural, because demand is served by an ED across the border.
- Overdispersion: attendance rates vary far more than population and age can explain.
- Space: neighbouring areas share EDs, GP practices and patient flows, so their propensities move together.

The question for the data

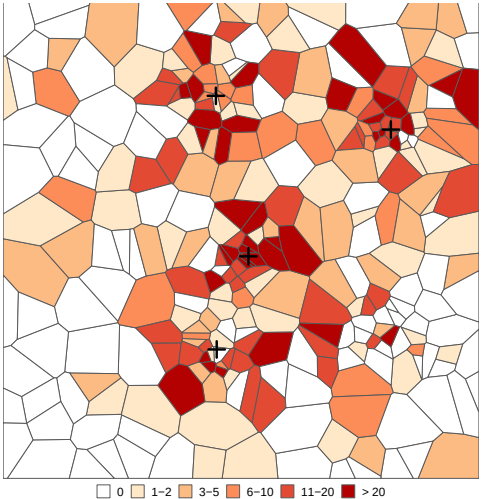
Does ED demand spill over between neighbouring areas — and which zeros are structural?

Plan of the workshop (50 minutes)

1. **The data problem** — spatial counts with excess zeros 6'
2. **Three toolkits that do not talk to each other** 9'
3. **The ZISAR model** — SAR on a latent field 12'
4. **Estimation** — Laplace + automatic differentiation (TMB) 8'
5. **Does it work?** Monte Carlo evidence 7'
6. **Interpretation** — direct, indirect and total effects 3'
7. **Open problems** and discussion 3'

1. The data problem

What the data look like



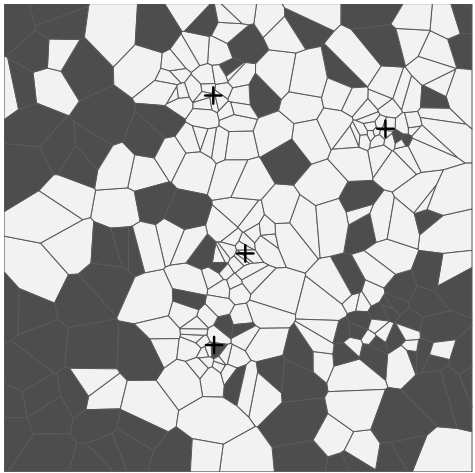
One municipality, one count.

300 municipalities, the number of emergency-department visits in a period. Four hospitals with an ED, marked +.

Big counts around the hospitals, small counts in the periphery — and a lot of white.

What drives them: population and distance to the nearest ED (and distance also drives the zeros). With real data: age structure, deprivation, GP supply.

What the data look like



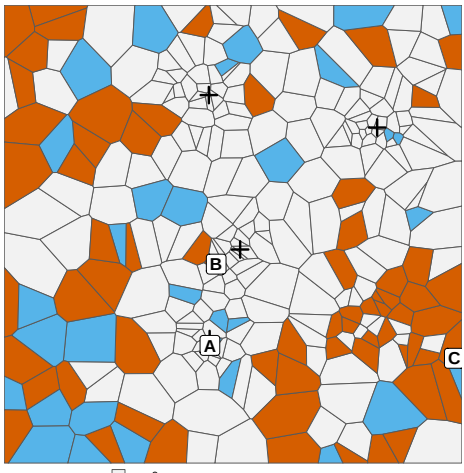
Feature 1 — many zeros.

One municipality in three reports *nothing*.

A Poisson with the right mean predicts almost none of them. A negative binomial can get there, but only by distorting every other parameter.

And the zeros are not scattered at random: they sit together.

What the data look like



Feature 2 — two kinds of zero.

Sampling zero: the municipality is "on", but its risk is low and nothing happened this period.

A — a small village of young families next to a hospital: nobody needed the ED this month; next month it may report two visits.

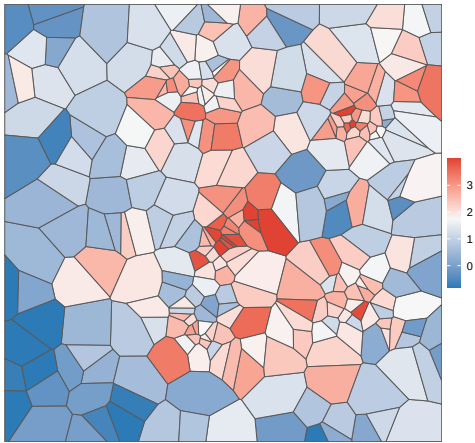
Structural zero: no demand can reach these EDs.

B — an industrial area or a nature reserve next to a hospital: almost nobody lives there.

C — a border municipality: residents use the hospital of the next region, outside our data.

Same zero on the map; a different regime underneath.

What the data look like



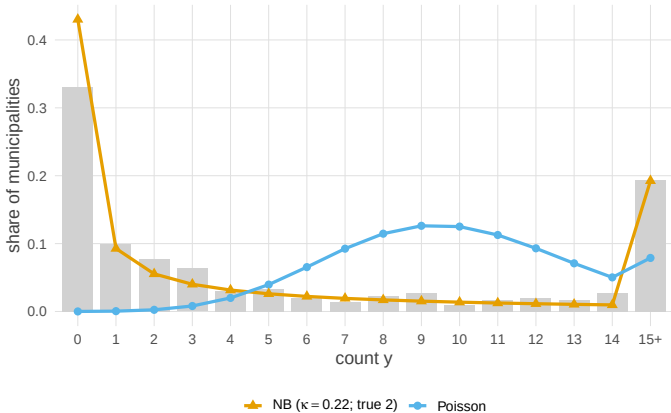
Feature 3 — neighbours look alike.

Behind every count there is an *underlying risk* of using the ED (the model calls it the propensity η_i). Take away the zeros and the luck of the draw: this map is what is left.

Neighbouring municipalities have similar risk — partly for observed reasons (population, distance), partly for unobserved ones: the same GP practice, the same habits, the same ambulance service.

ZISAR estimates this map; ρ says how strongly a municipality's risk follows its neighbours'.

Three features, one signature



1 Many zeros

33% observed. A Poisson predicts none; an NB gets there (and beyond, 43%) only with $\kappa = 0.22$ (true 2). And *two kinds*: sampling, structural.

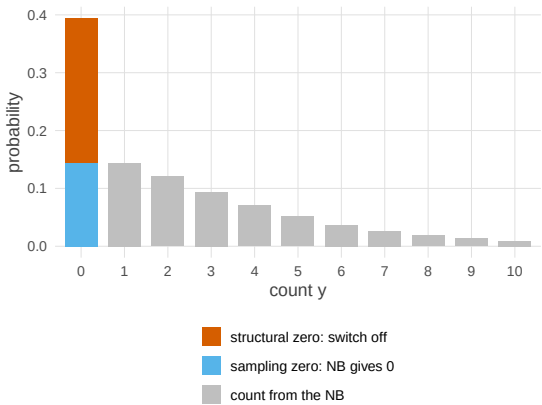
2 Overdispersion

variance $\gg$ mean among the non-zeros; a Poisson understates every standard error.

3 Spatial dependence

neighbours look alike even after covariates.

How a zero-inflated negative binomial works



Two steps for each municipality

1. **A switch.** With probability π_i the process is off: structural zero.
2. **A count.** Otherwise $y_i \sim \mathrm{NB}(\mu_i, \kappa)$, which can itself give a sampling zero.

$$P(y_i = 0) = \pi_i + (1 - \pi_i)\,\mathrm{NB}(0)$$
$$P(y_i = k) = (1 - \pi_i)\,\mathrm{NB}(k)$$

Two regressions

- logit $\pi_i = \mathbf{z}_i'\boldsymbol{\gamma}$: who is switched off (distance to the ED).
- $\log \mu_i = \mathbf{x}_i'\boldsymbol{\beta}$: how many visits when on (population).
- NB variance $\mu_i + \mu_i^2/\kappa$: small κ, more spread than a Poisson.

2. Three toolkits that do not talk to each other

Toolkit A — spatial econometrics: the SAR model

$$\mathbf{y} = \rho\,\mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \qquad \Longrightarrow \qquad \mathbf{y} = \underbrace{(\mathbf{I} - \rho\mathbf{W})^{-1}}_{\mathbf{S}(\rho)}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})$$

- Anselin (1988), Lee (2004), Kelejian–Prucha (1998, 99), LeSage–Pace (2009), Arbia (2006, 2024).
- ρ is a *structural* parameter: endogenous interaction, not residual correlation.
- $\mathbf{S}(\rho) = \sum_s \rho^s \mathbf{W}^s$ gives the direct / indirect / total effects decomposition — the object policy people actually want.
- Built for continuous $\mathbf{y}$. For counts the lag moves into the mean: on $\log \boldsymbol{\mu}$ (Lambert, Brown & Florax 2010, two-step LIML) or on a latent Gaussian field (Liesenfeld, Richard & Vogler 2016, likelihood by importance sampling). Both Poisson, neither handles excess zeros.

Toolkit B — biostatistics: zero-inflated counts with spatial random effects

$$\text{logit}(\pi_i) = \mathbf{z}_i'\boldsymbol{\gamma} + g_i, \qquad \log \mu_i = \mathbf{x}_i'\boldsymbol{\beta} + v_i, \qquad (\mathbf{g}, \mathbf{v}) \text{ CAR / Gaussian process}$$

- Besag–York–Mollié (1991) random effects inside a ZIP/ZINB; `CARBayes` (Lee 2013), Neelon et al. (2013), INLA-based disease mapping, NNGP (He et al. 2024).
- Usually Bayesian; excellent at smoothing and mapping risk.
- Spatial dependence is a latent nuisance, not an interaction: no ρ with an interaction meaning, no indirect effects, no "what happens in i if I intervene in j".

Toolkit C — spatial filtering

- **Idea** (Griffith 2003): add to the regression a few "map patterns" — eigenvectors of **W** — that soak up the spatial correlation, then fit any ordinary ZIP/ZINB.
- Used inside a zero-inflated model by Metulini, Patuelli & Griffith (2018).
- Pragmatic and easy. But there is no ρ and no spillover: space is removed, not measured.

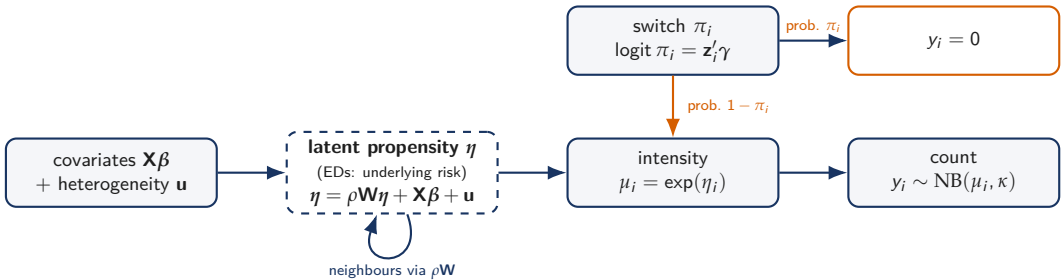
The gap

Model	Spatial lag ρ	Zero-infl.	Overdisp.	Heterog.	Spillovers
SAR-Poisson (Lambert et al. 2010)	✓	–	–	–	✓
SAL-Poisson (Liesenfeld et al. 2016)	✓	–	–	unit-level	✓
ZINB (Cameron–Trivedi 2013)	–	✓	✓	–	–
ZIP + CAR (Lee 2013; Neelon et al. 2013)	–	✓	–	CAR	–
ZINB + NNGP (He et al. 2024)	–	✓	✓	GP	–
ZIP + eigenvector filtering	(proxy)	✓	–	–	–
ZISAR	✓	✓	✓	unit-level	✓

What is new: a structural ρ inside a ZINB, a fast full likelihood (Laplace + TMB), and evidence on what goes wrong without it.

3. The ZISAR model

The idea in one picture



- **In words:** each place has a latent propensity η_i — for EDs, its underlying risk of attendance. Propensities interact across neighbours *before* they are turned into counts. On top, a switch can set the count to zero.
- **The design choice:** spatial feedback acts on $\boldsymbol{\eta}$, not on the observed $\mathbf{y}$.

Formally

$$y_i \sim \begin{cases} 0 & \text{w.p. } \pi_i \\ \mathrm{NB}(\mu_i, \kappa) & \text{w.p. } 1-\pi_i \end{cases} \qquad \mathrm{Var}[y_i \mid \text{count}] = \mu_i + \mu_i^2/\kappa$$

$$\mu_i = \exp(\eta_i), \qquad \boldsymbol{\eta} = \rho\mathbf{W}\boldsymbol{\eta} + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{u} \sim N(\mathbf{0}, \sigma^2\mathbf{I}) \qquad \text{SAR on the latent predictor}$$

$$\boldsymbol{\eta} = \mathbf{S}(\rho)\,(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}), \qquad \mathbf{S}(\rho) = (\mathbf{I} - \rho\mathbf{W})^{-1} \qquad \text{reduced form}$$

$$\mathrm{logit}(\pi_i) = \mathbf{z}_i'\boldsymbol{\gamma} \qquad \text{the switch}$$

Parameters: $\boldsymbol{\theta} = (\boldsymbol{\beta}', \rho, \boldsymbol{\gamma}', \kappa, \sigma^2)'$.

Implied covariance of the latent field

$\mathrm{Var}[\boldsymbol{\eta}] = \sigma^2\,\mathbf{S}(\rho)\mathbf{S}(\rho)' = \sigma^2\left[(\mathbf{I} - \rho\mathbf{W})'(\mathbf{I} - \rho\mathbf{W})\right]^{-1}$ — the SAR covariance of Anselin (1988), to be contrasted with the CAR precision $(\mathbf{I} - \rho\mathbf{W})$ of disease mapping.

The switch has no space: what setting $g_i = 0$ means

$$\text{logit}(\pi_i) = \mathbf{z}_i'\boldsymbol{\gamma} + g_i, \qquad \mathbf{g} = (\mathbf{I} - \rho_0 \mathbf{W})^{-1}\boldsymbol{\varepsilon}_0 \qquad \longrightarrow \qquad \text{today: } g_i = 0$$

- **What it says.** Structural zeros cluster only as much as their covariates do. Two border villages are both switched off because both are far from an ED — not because of each other.
- **What it rules out.** Zeros that cluster for reasons **Z** does not see: a valley served by the hospital of the next region. If that happens, the model has one place to put the pattern — low intensity in the count field — so it can leak into $\hat{\rho}$ and into the split between structural and sampling zeros.
- **What it implies for effects.** Covariates in the switch act only locally: no spillover through the zeros.
- **Why we accept it.** The general model adds a second latent field of size n and two parameters (ρ_0, σ_0^2), and ρ_0 is learnt from the zeros alone: weakly identified unless they are many. Not yet in our code.
- **How to check it.** Choose **Z** with a substantive story (distance, catchment), and look for clustering in the switch residuals. The general model is the next step.

Why SAR on η and not on y?

SAR on $\mathbf{y}$: "my count depends on your count"

$\log \mu_i = \mathbf{x}_i'\boldsymbol{\beta} + \rho \sum_j w_{ij}\, y_j + v_i$

- y_j is on both sides $\Rightarrow$ endogenous
- counts have no reduced form $\Rightarrow$ instruments, no likelihood

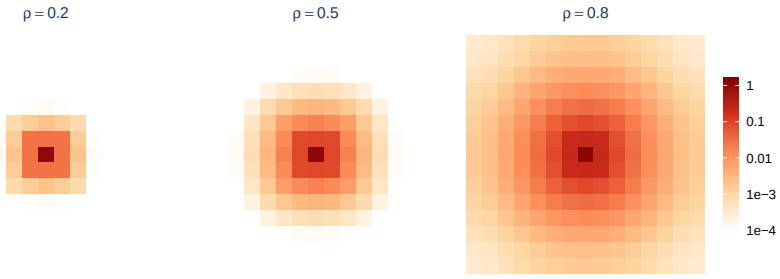
SAR on $\boldsymbol{\eta}$: "my risk depends on your risk"

$\boldsymbol{\eta} = \rho \mathbf{W}\boldsymbol{\eta} + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$

- $\boldsymbol{\eta} = \mathbf{S}(\rho)(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) \Rightarrow$ full likelihood, no instruments
- same $\mathbf{S}(\rho) \Rightarrow$ same direct / indirect effects as the linear SAR

Lambert et al. (2010) and Liesenfeld et al. (2016) already put the lag here, for Poisson counts. ZISAR adds the zeros.

What ρ does: the spatial multiplier



- Something raises ED demand in the central municipality — a care home opens, a GP practice closes. The colour is how much of that shock reaches each municipality around it (log scale).
- $\rho = 0.2$: only the immediate neighbours feel it. $\rho = 0.5$: three or four rings. $\rho = 0.8$: the whole region.
- Mechanically, $\mathbf{S}(\rho) = \mathbf{I} + \rho\mathbf{W} + \rho^2\mathbf{W}^2 + \cdots$: neighbours, then neighbours of neighbours, each time weaker.

ZISAR contains its competitors

Switch off	You get
the zeros ($\pi_i = 0$)	SAR-NB, a spatial negative binomial
space ($\rho = 0,\ \sigma^2 = 0$)	ZINB
zeros and extra variance ($\pi_i = 0,\ \kappa \to \infty,\ \sigma^2 = 0$)	SAR-Poisson (Lambert et al. 2010)
everything	Poisson regression

- So we can ask the data which pieces they need: fit with and without, and compare (a likelihood-ratio test).

4. Estimation

The likelihood, and how we compute it

$$\ell(\boldsymbol{\theta}) = \log \int_{\mathbb{R}^n} \underbrace{\prod_{i=1}^{n} f(y_i \mid \eta_i, \pi_i; \kappa)}_{\text{ZINB for the counts}} \quad \underbrace{p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma^2)}_{\text{SAR for the latent map}} \quad \mathrm{d}\boldsymbol{\eta}$$

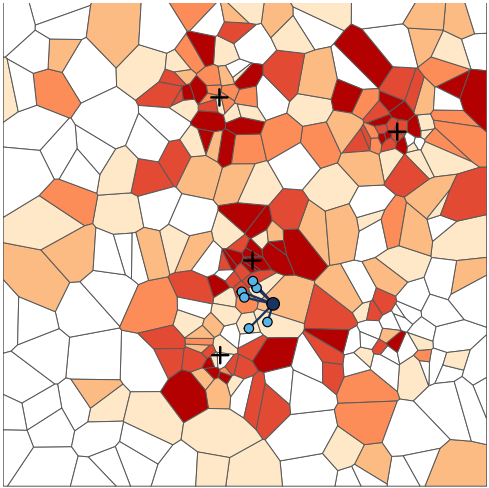
- In words: average the fit of the counts over every latent risk map the SAR finds plausible — an integral in 300 dimensions, with no closed form.

How: Laplace + TMB

1. For given parameters, find the most plausible latent map $\hat{\boldsymbol{\eta}}$. A few Newton steps, cheap because each municipality only talks to its 6 neighbours.
2. Approximate the integral by a Gaussian centred there (Laplace approximation).
3. TMB (Template Model Builder, Kristensen et al. 2016) computes all derivatives automatically; an optimiser adjusts the 8 parameters. **0.14 s** for 300 municipalities.

Alternatives: MCMC (slow), importance sampling (Liesenfeld et al.), INLA (not built for a SAR with a free ρ).

Estimation on the map, step by step



Step 0 — the data.

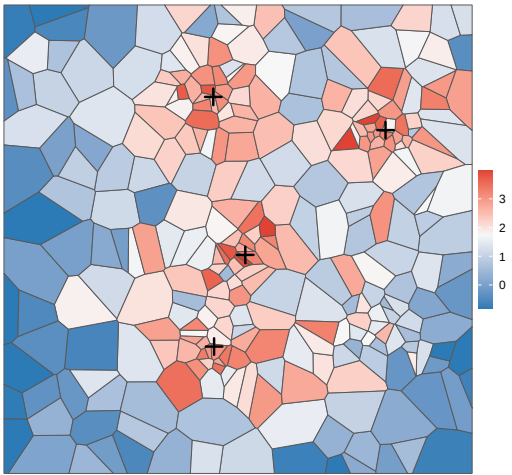
300 municipalities, 99 zeros (33%), mean 9.9 visits, variance 456.

Covariates: population and distance to the nearest ED; distance also enters the switch.

Neighbours: the 6 nearest municipalities, each with weight $1/6$ (lines on the map).

Unknowns: 8 parameters $(\boldsymbol{\beta}, \rho, \gamma, \kappa, \sigma)$ and the 300 latent risks η_i.

Estimation on the map, step by step

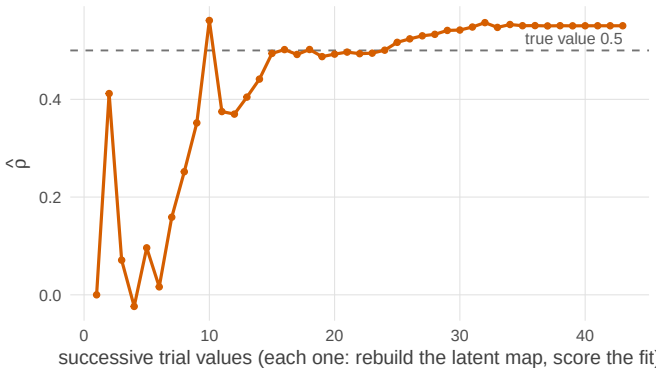


Step 1 — a starting guess.

A plain Poisson regression on population and distance: the risk map the covariates alone can explain. No space, no zeros yet.

For comparison, an aspatial NB on the same data finds a distance effect of -0.92 (truth -0.20) and $\kappa = 0.64$ (truth 2): it pushes the zeros and the clusters into the coefficients.

Estimation on the map, step by step



Step 2 — the search.

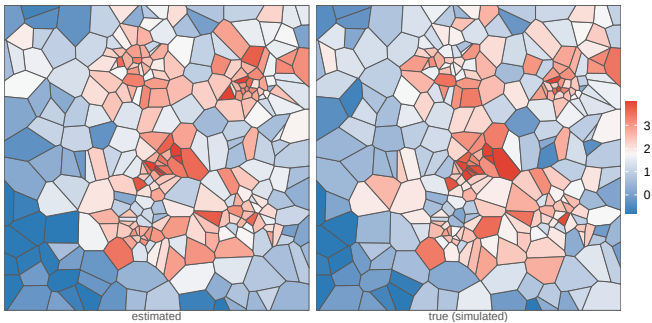
The optimiser tries parameter values. For each trial it

- rebuilds the most plausible latent risk map: zeros pull a municipality down, large counts pull it up, neighbours pull each other together;
- scores how well that map explains the counts.

$\hat{\rho}$ starts at 0, overshoots, settles at 0.55.

34 iterations, **0.14 s**. TMB supplies exact derivatives, so the search is short.

Estimation on the map, step by step



Step 3 — the result.

	truth	est.	SE
$\beta_{\text{population}}$	0.70	0.77	0.08
β_{distance}	−0.20	−0.19	0.07
ρ	0.50	0.55	0.08
γ_{distance}	1.40	1.42	0.26
κ	2.00	1.38	0.32
σ	0.50	0.37	0.14

Is there spatial dependence?
Likelihood-ratio test of $\rho = 0$: 18.5, $p \approx 2 \cdot 10^{-5}$.
Estimated vs true risk map: correlation 0.91.

κ and σ both below truth: two ways of saying "overdispersion" that trade off — the one weak spot in the Monte Carlo too.

5. Does it work? Monte Carlo

Design

DGP — restricted ZISAR, queen contiguity on a square lattice

- $n \in \{100, 225, 400\}$, $\rho \in \{0, 0.15, 0.3, 0.45, 0.6\}$, $R = 1000$ per cell
- $\boldsymbol{\beta} = (0.5, 0.8, -0.5)'$, $\boldsymbol{\gamma} = (-1, 0.4)'$, $\kappa = 2$, $\sigma = 0.5$
- $\approx$ 40–50% zeros, structural + sampling

Competitors

- **ZINB** — `pscl::zeroinfl`, ignores space
- **SAR-NB** — NB GLM with instrumented $\widehat{\mathbf{Wy}}$, $\mathbf{H} = [\mathbf{X}, \mathbf{WX}, \mathbf{W}^2\mathbf{X}]$, ignores zeros
- **ZISAR** — TMB / Laplace

The ZINB has no ρ: it assumes $\rho = 0$, so in the $\hat{\rho}$ plots it sits on zero.

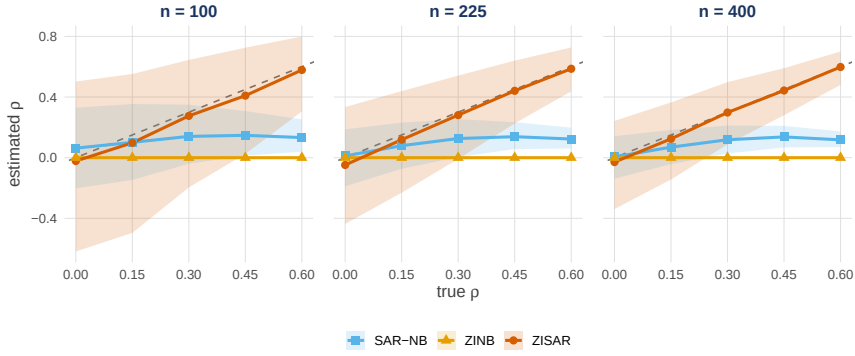
What we look at

- the estimated $\hat{\rho}$ and $\hat{\beta}_1$ against the true values
- RMSE, convergence, seconds per fit

The two questions

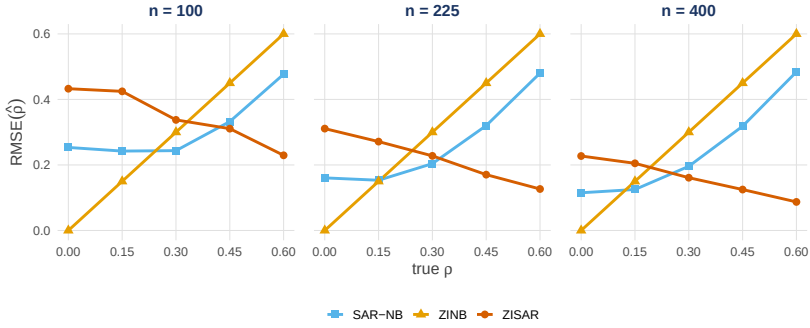
1. What does ignoring zeros do to $\hat{\rho}$?
2. What does ignoring space do to $\hat{\boldsymbol{\beta}}$?

Ignoring the zeros destroys $\hat{\rho}$ — and n does not help



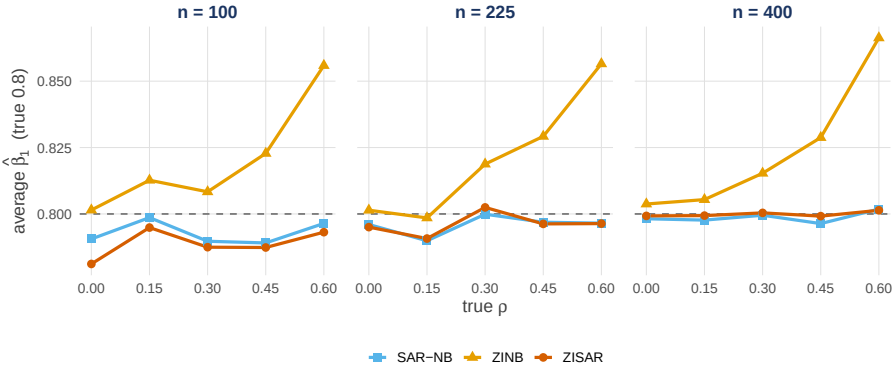
- Lines: average estimate over 1000 replications; bands: 10th–90th percentile; dashed: the truth. The ZINB sits at 0 by construction.
- SAR-NB flattens out: at $\rho_0 = 0.6$ it estimates 0.13, 0.12, 0.12 as n grows. Not a small-sample problem: the model is wrong.
- ZISAR follows the diagonal (0.58, 0.59, 0.60 at $\rho_0 = 0.6$); its band is wide at $n = 100$ and narrows fast.

Precision of $\hat{\rho}$



- At $\rho_0 = 0$ the SAR-NB is tighter (RMSE 0.12 vs 0.23 at $n = 400$): with nothing to find, a simpler estimator wins. ZISAR pays for the extra variance component.
- From $\rho_0 \approx 0.3$ (0.45 at $n = 100$) the ranking reverses, and the SAR-NB error is essentially its bias (0.48 at $\rho_0 = 0.6$, vs 0.09 for ZISAR).
- Honest note: 10% of ZISAR fits (17% at $n = 100$) fail to converge or put $\hat{\kappa}$ at the boundary — the κ vs σ^2 trade-off — and are excluded.

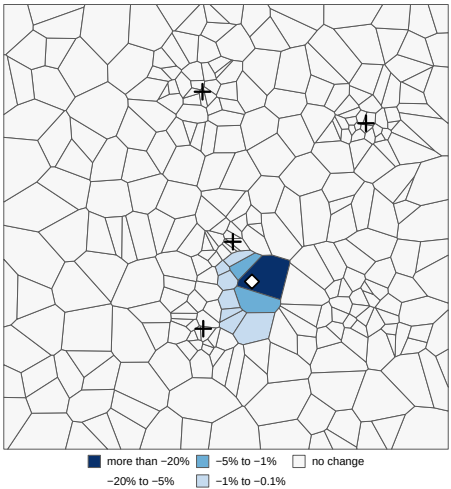
Ignoring space biases $\hat{\beta}$ and costs efficiency



- The ZINB drifts away from 0.8 as ρ_0 grows ($+0.07$ at $\rho_0 = 0.6$), and more data do not help: the omitted spatial field ends up in the coefficient.
- ZISAR and SAR-NB stay on the truth. At $\rho_0 = 0.6$, $n = 400$: RMSE($\hat{\beta}_1$) 0.079 ZISAR, 0.090 SAR-NB, 0.111 ZINB. At $\rho_0 = 0$ all three coincide.

6. Interpretation

Coefficients are not effects: direct and indirect



An urgent-care clinic opens in the municipality marked ◇; it lowers its ED risk (η) by 0.3. On the map of slide 5, with $\rho = 0.5$:

- **Direct effect**: -1.6 expected ED visits there (-26%), including a small echo back from the neighbours.
- **Indirect effect** (spillover): -0.7 visits spread over 9 neighbouring municipalities, where nothing changed.
- **Total**: -2.3 visits for the region. A third of the effect happens next door.

A ZINB reports only the direct effect. Averaged over all municipalities, with $\rho = 0.5$ the total is twice the direct.

7. Open problems

What I would like to discuss with you

Methodological

- Identification of ρ vs ρ_0 in the general model (spatial switch): when do zeros carry their own spatial signal?
- σ^2 vs κ^{-1}: two ways to say "overdispersion"; boundary estimates in 10–15% of fits.
- Laplace accuracy with very sparse counts; higher-order corrections or importance sampling on top of TMB.
- $O(n^3)$ eigenvalues of $\mathbf{W}$: stochastic log-determinants, or the pairwise/KD-tree ideas of Arbia–Nardelli–Salvini.

Extensions

- Space–time: $\eta_{it} = \rho(\mathbf{W}\boldsymbol{\eta}_t)_i + \phi\,\eta_{i,t-1} + \mathbf{x}_{it}'\boldsymbol{\beta} + u_i + \nu_t$.
- Several $\mathbf{W}$'s (geographic, economic, network proximity) — the "relational" side of territorial capital.
- Other count layers (CMP, generalised Poisson), other links for the switch.
- An `R` package with formula interface and `spdep`/`sf` integration.

Applied

- Which regional outcomes in your work look like this? Firm births, patents, migration flows at small scale?

Limitations, stated plainly

1. **Asymptotics.** One latent η_i per observation: the textbook $O(n^{-1})$ Laplace error is for a fixed number of random effects. Our consistency argument needs the approximation error to be negligible per unit; the SAR prior helps, but this is a condition we are still verifying. Today the evidence is the Monte Carlo.
2. **Zeros vs. low intensity.** A zero can come from the switch or from a very low η_i; with a free u_i per unit the switch is identified mainly through $\mathbf{Z}$ and the functional form. Choose $\mathbf{Z}$ with a substantive story.
3. κ **vs.** σ^2**.** Two ways of saying "overdispersion"; in small samples the split is weakly identified (10–15% of Monte Carlo fits at the boundary).
4. **Competitors and data.** The Monte Carlo does not yet include a Bayesian CAR competitor nor ZISAR with $\pi = 0$; the application still runs on synthetic data, real data next (York).

Take-aways

1. Spatial counts with excess zeros need both a switch and a spatial interaction; existing tools give you one or the other.
2. Putting the SAR on the latent propensity keeps the econometric meaning of ρ and the effects decomposition, and makes maximum likelihood feasible.
3. Laplace + automatic differentiation turns an n-dimensional integral into a one-second fit.
4. Ignoring zeros flattens $\hat{\rho}$ (0.12 when the truth is 0.6) and more data do not help; ignoring space biases $\hat{\boldsymbol{\beta}}$ and reports zero spillovers.

Thank you.

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Backup

Backup: illustrative application on a Piedmont-like region

Setting

- $n = 300$ municipalities, all-cause ED visits in one period (synthetic, Piedmont-like)
- 25% zeros; mean 6.1, variance 81
- **W**: 6 nearest neighbours, row-standardised

Count X: log-population, aging index, distance to nearest ED, income

Switch Z: distance to nearest ED, GP density

Why these two. A structural zero is "no demand reaches this ED": distance, and GPs that absorb demand.

What the synthetic run shows (truth: $\rho = 0.35$, $\kappa = 2$)

- **ZISAR**: $\hat{\rho} = 0.41$ (SE 0.15), $\hat{\kappa} = 2.03$, $\hat{\sigma} = 0.39$; every $\hat{\beta}$ within one SE of the truth.
- **SAR-NB** (no zeros): the distance effect vanishes (-0.03 vs. 0.20) and $\hat{\kappa} = 0.69$: the zeros become overdispersion.
- **ZINB** (no space): slopes close, but the intercept absorbs the multiplier (1.92 vs. 1.20) and $\hat{\kappa} = 1.33$.
- Indirect $\approx 40\%$ of total, imprecise at $n = 300$.
- Moran's $I \approx 0$ and n.s. on all three residuals — and on **y** itself (0.02, $p = 0.24$). With zeros and overdispersion a residual test misses a dependence the likelihood estimates at $\hat{\rho} = 0.41$.

Next: real data

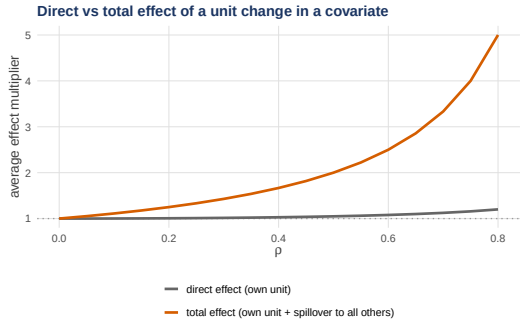
April–May 2027, CHE York, with Rita Santos: ED attendances by small area (slide 2).

Backup: the effects decomposition, formally

Change covariate r in unit j; the response of $\mathbb{E}[\mathbf{y}]$ in every unit i is the $n \times n$ matrix

$$\mathbf{M}_r = \operatorname{diag}\big((\mathbf{1} - \hat{\pi}) \odot \hat{\mu}\big)\, \mathbf{S}(\rho)\, \beta_r \;+\; \underbrace{\big[-\operatorname{diag}(\hat{\mu})\operatorname{diag}(\hat{\pi} \odot (\mathbf{1} - \hat{\pi}))\, \gamma_r\big]}_{\text{inflation channel, diagonal if } r \in \mathbf{Z}}$$

$$\overline{M}_r^{\text{direct}} = n^{-1} \operatorname{tr} \mathbf{M}_r, \qquad \overline{M}_r^{\text{total}} = n^{-1} \mathbf{1}' \mathbf{M}_r \mathbf{1}, \qquad \overline{M}_r^{\text{indirect}} = \text{total} - \text{direct}.$$



- LeSage–Pace (2009) scaled by the ZINB mean $(1 - \pi_i)\mu_i$.
- Indirect effect $= 0$ *iff* $\rho = 0$; sign of β_r preserved; monotone in $|\rho|$.
- SEs by simulating $\boldsymbol{\theta}^{(b)} \sim N(\hat{\boldsymbol{\theta}}, \hat{\mathbf{V}})$, $B = 500$.
- A ZINB reports zero spillover by construction.

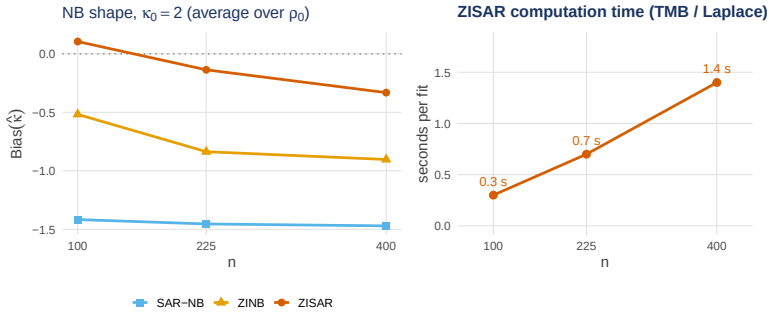
Backup: asymptotics (the short version)

Assumptions

(W) $\mathbf{W}_n$ uniformly bounded in row/column sums, ρ in a compact subset of $(-1/r_{\max}, 1/r_{\max})$; (X) exogenous regressors with $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{Q}_X > 0$; (P) true $\boldsymbol{\theta}_0$ interior, $\kappa_0, \sigma_0^2 > 0$; (I) $\mathbb{E}[\ell(\boldsymbol{\theta})]$ uniquely maximised at $\boldsymbol{\theta}_0$.

- **Consistency.** Laplace error is $O(n^{-1})$ (Tierney–Kadane 1986), so $n^{-1}\ell_{\mathrm{LA}} \to n^{-1}\ell$ uniformly; standard extremum-estimator argument (Newey–McFadden 1994).
- **Asymptotic normality.** $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)^{-1})$ by mean-value expansion of the score and a CLT for spatially dependent arrays under (W) — the Lee (2004) route.
- **Honest caveat.** The Laplace approximation is a first-order approximation to a non-Gaussian integral; with very sparse counts and many zeros its accuracy is the thing to check (we do, in the Monte Carlo).

Backup: overdispersion attribution, fit and speed ($R = 100$)



- Both competitors push κ down ("more overdispersion") to explain zeros and clusters they do not model; SAR-NB is off by -1.4 on a true value of 2.
- Median log-likelihood: ZISAR best whenever $\rho_0 > 0$, gap over ZINB growing with ρ_0 and n (15 nats at $n = 400$, $\rho_0 = 0.6$: -869 vs -884); indistinguishable at $\rho_0 = 0$.
- Convergence 96–100%; 10–15% of fits hit the κ boundary (the classic σ^2 vs κ^{-1} trade-off) and are flagged.

Backup: the estimation algorithm

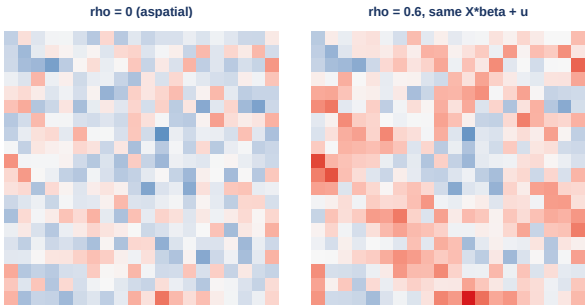
1. **Pre-compute** eigenvalues of $\mathbf{W}$; sparse Cholesky ordering of $(\mathbf{I}-\rho\mathbf{W})'(\mathbf{I}-\rho\mathbf{W})$.
2. **Initialise** $\boldsymbol{\beta}^{(0)}$ from a Poisson GLM, $\gamma^{(0)} = (\text{logit}(\text{share of zeros}), 0, \dots)$, $\rho = 0$, $\kappa = 1$, $\sigma = 0.5$; $\boldsymbol{\eta}^{(0)} = \mathbf{X}\boldsymbol{\beta}^{(0)}$.
3. **Objective**: $f(\boldsymbol{\eta};\boldsymbol{\theta}) = -\sum_i \log f(y_i \mid \eta_i, \pi_i; \kappa) - \log p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma^2)$, with $\boldsymbol{\eta}$ declared as random effects in TMB.
4. **Repeat**: inner Newton–Raphson for $\hat{\boldsymbol{\eta}}(\boldsymbol{\theta}^{(s)})$ with sparse Cholesky; evaluate ℓ_{LA} and $\nabla_{\boldsymbol{\theta}}\ell_{\text{LA}}$ by AD; `nlminb` step on $\boldsymbol{\theta}$.
5. **Output**: $\hat{\boldsymbol{\theta}}$, AD Hessian $\Rightarrow \hat{\mathbf{V}}$, delta-method SEs on the original scale (`sdreport`).

ZINB pmf with mean e^{η} and shape κ

$$f(y \mid \eta, \pi; \kappa) = \begin{cases} \pi + (1-\pi)\, h(0; \eta, \kappa) & y = 0 \\ (1-\pi)\, h(y; \eta, \kappa) & y \geq 1 \end{cases}$$

$$h(y; \eta, \kappa) = \frac{\Gamma(y+\kappa)}{\Gamma(\kappa)\Gamma(y+1)} \left(\frac{\kappa}{\kappa + e^{\eta}}\right)^{\kappa} \left(\frac{e^{\eta}}{\kappa + e^{\eta}}\right)^{y}.$$

Backup: spatial dependence is not just a nuisance



Same covariates, same unobserved heterogeneity. The only difference is a feedback parameter ρ.

- Left: what an aspatial model assumes.
- Right: what a spatial-lag model generates — clusters that are *not* in $\mathbf{X}$.
- An aspatial regression on the right panel attributes the clusters to whatever covariate happens to be clustered too.

This is the mechanism behind the bias we will see in the Monte Carlo.

Backup: zero-inflation or hurdle?

Zero-inflated (ZISAR)

$$P(y{=}0) = \pi + (1-\pi)h(0), \quad P(y{=}k) = (1-\pi)h(k)$$

- two sources of zeros, mixed: a unit can be "on" and still record zero
- the switch and the count share the zeros, so π and μ are identified jointly through the covariates and the spatial structure
- natural when the structural zero is a latent state (no at-risk population, demand served elsewhere)

Hurdle (Neelon et al. 2013)

$$P(y{=}0) = \pi, \quad P(y{=}k) = (1-\pi)\,\frac{h(k)}{1-h(0)}$$

- all zeros from the first stage; the count is truncated at zero
- two likelihoods that separate cleanly — simpler, but the count process no longer "owns" any zero
- natural when zero is an observed decision (participate / not)

In the TMB template

The hurdle version changes the observation block only; latent field, Laplace step and effects are unchanged. Both can be fitted and compared by AIC/BIC on the same data.

Backup: three channels of overdispersion, and how they are told apart

Marginal variance of y_i (integrating the latent field), with $\mathbb{E}[\mu_i] = \exp\!\big(\bar{\eta}_i + \tfrac{1}{2}\sigma^2[\mathbf{SS}']_{ii}\big)$:

$$\mathrm{Var}[y_i] = (1-\pi_i)\,\mathbb{E}[\mu_i]\big(1+\kappa^{-1}\mathbb{E}[\mu_i]\big) + \pi_i(1-\pi_i)\,\mathbb{E}[\mu_i]^2 + (1-\pi_i)\;\mathrm{Var}[\mu_i]$$

- κ **(NB shape)**: extra variance *within* a unit, proportional to μ_i^2, no spatial pattern.
- π_i **(zeros)**: extra variance from the mixture, tied to $\mathbf{Z}$ and to the share of zeros.
- σ^2 **(heterogeneity)**: extra variance *between* units, amplified by $[\mathbf{S}(\rho)\mathbf{S}(\rho)']_{ii}$ — so it is spatially patterned when $\rho \neq 0$.

Identification in practice

κ and σ^2 are separated by the *shape* of the extra variance (quadratic in μ vs log-normal) and by its spatial structure; π_i by the excess at zero. Where the data are too sparse to separate κ from σ^2, the fit drifts to the κ boundary — the 10–15% of Monte Carlo replications we flag. A profile likelihood in $\log\kappa$ makes this visible for a given dataset.

Backup: how good is the Laplace approximation here?

- The integrand in $\boldsymbol{\eta}$ is a Gaussian SAR prior times a product of ZINB terms; each term is log-concave in η_i for $y_i \geq 1$, and for $y_i = 0$ it is bounded and smooth. The conditional mode is unique in all our fits.
- Laplace is exact for Gaussian observation models and has error $O(n^{-1})$ in the marginal likelihood (Tierney–Kadane 1986); the practical risk is *skewness* of $p(\eta_i \mid y_i)$ when $y_i = 0$ and π_i is large.
- **Checks we can run on any dataset:** (i) importance sampling around the Laplace mode (TMB: `MCMC` / `tmbstan` for a Hamiltonian check); (ii) refit with the restricted vs general switch; (iii) compare with `glmmTMB` ZINB + i.i.d. RE at $\rho = 0$, which uses the same approximation.
- **Monte Carlo evidence:** bias of $\hat{\rho}$ and $\hat{\boldsymbol{\beta}}$ is negligible at $n \geq 225$ with 40–50% zeros, which is the regime where the approximation would be expected to struggle.
- Where it is not enough: higher-order Laplace (Shun–McCullagh), or INLA-style numerical integration over the few hyper-parameters with Laplace only for the latent field.

Backup: SAR on η vs CAR — what is the difference in practice?

SAR (ZISAR)

$$\mathrm{Var}[\boldsymbol{\eta}] = \sigma^2 \left[(\mathbf{I} - \rho \mathbf{W})' (\mathbf{I} - \rho \mathbf{W}) \right]^{-1}$$

- precision is quadratic in $\mathbf{W}$: second-order neighbours enter
- ρ is the coefficient of a simultaneous equation — an interaction
- effects: $\partial \mathbb{E}[\mathbf{y}] / \partial x_{jr}$ via $\mathbf{S}(\rho)$, so a shock in j reaches every i
- the mean $\mathbf{S}(\rho)\mathbf{X}\boldsymbol{\beta}$ is spatially multiplied too

CAR (BYM, CARBayes)

$$\mathrm{Var}[\boldsymbol{\eta}] = \sigma^2 \left(\mathbf{D} - \rho \mathbf{W} \right)^{-1}$$

- precision is linear in $\mathbf{W}$: only first-order neighbours
- ρ is a smoothing parameter of a conditional prior
- the mean is $\mathbf{X}\boldsymbol{\beta}$ unchanged: covariates do not spill over
- usually symmetric $\mathbf{W}$ required; row-standardised $\mathbf{W}$ needs the Leroux/proper-CAR variant

Both are Gaussian Markov random fields; the SAR is the one whose parameter has an econometric reading. In the TMB template, swapping the prior block gives the CAR version of the same zero-inflated model, which is a useful robustness check.

Backup: choosing W, and what the diagnostics look like

- **Candidates for municipalities:** queen contiguity; k-nearest neighbours ($k = 4, 6, 8$) on centroids; inverse-distance with a cut-off; travel-time to the nearest ED. Row-standardised so that $\rho \in (-1, 1)$ and $\mathbf{W}\boldsymbol{\eta}$ is a neighbourhood average.
- **Selection:** fit each, compare marginal log-likelihood / AIC (the log det term is already in the likelihood, so this is legitimate); report $\hat{\rho}$ under all of them — the sign and significance should be stable, the magnitude will not be (it scales with the density of $\mathbf{W}$).
- **Residual diagnostics:** Moran's I on Pearson residuals $(y_i - \hat{\mathbb{E}}[y_i]) / \sqrt{\widehat{\mathrm{Var}}[y_i]}$ under the chosen $\mathbf{W}$ *and* under the alternatives; randomised quantile residuals for the count layer; observed vs predicted share of zeros by neighbourhood.
- **In the synthetic Piedmont run:** Moran's I is ≈ 0 and not significant on all three models' residuals and on $\mathbf{y}$ itself (0.02, $p = 0.24$), while ZISAR estimates $\hat{\rho} = 0.41$ (SE 0.15): with zeros and overdispersion, residual tests have little power — use the likelihood (LR on ρ) as the main diagnostic.
- **Endogenous W** (e.g. patient flows) is a different model: it is on the extensions list, not in the current estimator.

Backup: computational cost and scaling

Step	Cost	Notes
Eigenvalues of $\mathbf{W}$ (once)	$O(n^3)$ dense	only step that does not scale; replace with Barry–Pace (1999) MC estimator or Chebyshev approximation
Inner Newton step	$O(n)$–$O(n^{3/2})$	sparse Cholesky of $\mathbf{H}$, fill-in depends on ordering (nested dissection for lattices)
$\log\det \mathbf{H}$	free	from the Cholesky factor
AD gradient of ℓ_{LA}	$\approx$ 2–4$\times$ one evaluation	reverse-mode AD through inner and outer problem
Outer `nlminb`	30–40 iterations	34 iterations on the 300-unit map, starting from a Poisson GLM

- Measured: 0.3 s ($n = 100$), 0.7 s ($n = 225$), 1.4 s ($n = 400$) per fit, single core, including `sdreport`.
- Not yet benchmarked beyond $n = 400$: the dense eigendecomposition is the first step to break as n grows; with a sparse $\log\det$ the rest scales with the Cholesky fill-in.
- The Monte Carlo: 9 scenarios $\times$ 100 replications $\times$ 3 models.

Backup: the general model with a spatial switch

$$\text{logit}(\pi_i) = \mathbf{z}_i'\gamma + g_i, \qquad \mathbf{g} = (\mathbf{I} - \rho_0 \mathbf{W})^{-1}\varepsilon_0, \quad \varepsilon_0 \sim N(\mathbf{0}, \sigma_0^2 \mathbf{I})$$

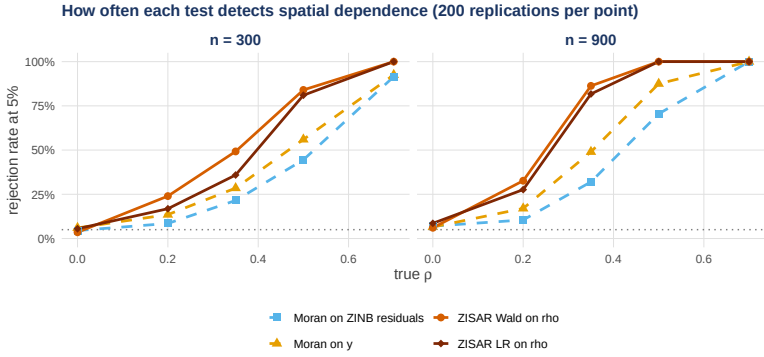
- **Why one might want it:** structural zeros cluster for their own reasons — a whole valley served by one hospital across the border, an area with no at-risk population. Then the switch is spatially organised independently of the count intensity.
- **What it costs:** a second latent field of dimension n in the Laplace approximation (same sparse machinery, roughly $2\times$ the cost), and two more parameters (ρ_0, σ_0^2).
- **Identification:** ρ_0 is identified only through the zeros, so it needs many of them and enough covariate variation in $\mathbf{Z}$; with 25% zeros and $n = 300$ the profile likelihood in ρ_0 is flat. With a shared field ($g_i \propto u_i$) identification is cleaner but the interpretation changes.
- **Effects:** the inflation channel of $\mathbf{M}_r$ is no longer diagonal, so covariates in $\mathbf{Z}$ also produce indirect effects.
- **Our position:** the restricted model is the workhorse; the general one is a sensitivity check and a likelihood-ratio test of "do the zeros cluster beyond what $\mathbf{Z}$ explains?"

Backup: the space–time extension

$$\eta_{it} = \rho\,(\mathbf{W}\boldsymbol{\eta}_t)_i + \phi\,\eta_{i,t-1} + \mathbf{x}_{it}'\boldsymbol{\beta} + u_i + \nu_t, \qquad y_{it} \sim \mathrm{ZINB}(\pi_{it},\, e^{\eta_{it}},\, \kappa)$$

- Same construction: a latent Gaussian field, now of dimension nT, with a sparse precision (block-tridiagonal in time, $\mathbf{W}$-sparse in space); TMB handles it with the same Laplace step.
- Stationarity requires $|\rho| + |\phi| < 1$ (for row-standardised $\mathbf{W}$); the long-run multiplier becomes $(\mathbf{I} - \rho\mathbf{W} - \phi\mathbf{I})^{-1}$, and effects split into short-run and long-run, direct and indirect.
- The unit effect u_i is now identified from the within-unit variation, which weakens the κ–σ^2 trade-off considerably.
- Open: initial conditions for η_{i0}; dynamic spatial panel asymptotics with fixed T; and whether the switch should have its own persistence.
- Natural datasets: monthly ED visits by municipality, yearly firm births by local labour market.

Backup: can a residual Moran test see the dependence?



- Same design as the application (6-NN, 25–35% zeros, $\kappa = 2$), 200 datasets per point, ZINB and ZISAR fitted on each.
- At $\rho = 0.35$, $n = 300$: Moran on ZINB residuals detects it 22% of the time, the ZISAR Wald test 49%. At $n = 900$: 32% vs. 86%.
- Under $\rho = 0$ all tests are close to 5%; the LR test is slightly liberal at $n = 900$ (8.6%).

Backup: Monte Carlo tables

Bias of $\hat{\rho}$ (RMSE in parentheses), $R = 100$

	$\rho_0 = 0$			$\rho_0 = 0.3$			$\rho_0 = 0.6$		
	100	225	400	100	225	400	100	225	400
SAR-NB	0.066 (.24)	−0.006 (.17)	0.018 (.11)	−0.170 (.26)	−0.180 (.20)	−0.184 (.20)	−0.464 (.47)	−0.474 (.48)	−0.479 (.48)
ZISAR	−0.019 (.39)	−0.037 (.34)	−0.015 (.23)	−0.105 (.42)	−0.032 (.20)	0.002 (.15)	−0.062 (.28)	−0.000 (.12)	−0.007 (.09)

Bias of $\hat{\beta}_1$ ($\beta_1^0 = 0.8$) and median log-likelihood

		$\rho_0 = 0$			$\rho_0 = 0.3$			$\rho_0 = 0.6$		
		100	225	400	100	225	400	100	225	400
Bias $\hat{\beta}_1$	ZINB	0.018	−0.009	0.006	−0.003	−0.000	0.009	0.053	0.059	0.055
	SAR-NB	0.009	−0.017	0.004	−0.018	−0.011	−0.009	0.016	0.002	0.000
	ZISAR	−0.012	−0.014	0.008	−0.020	−0.008	−0.003	0.011	−0.002	−0.002
Median ℓ	ZINB	−161	−363	−648	−176	−399	−716	−221	−494	−884
	SAR-NB	−162	−367	−655	−178	−402	−720	−224	−495	−887
	ZISAR	−160	−363	−648	−175	−396	−713	−218	−486	−869

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